Example 1 - Single Doc

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[1i] Summary

This rivt example calculates the maximum stress and deflection in a simply supported, uniformly loaded beam. It also serves as an annotated example of a *single doc* with multiple sections (a *single doc* does not use the report generating script).

The example illustrates the use of some of the most common API functions, commands and tags. Further details are provided in the [U] rivt user manual, https://www.rivt.info|.

The file may be formatted as a text, PDF or HTML doc by changing the type parameter in the PUBLISH command of the *Doc API (rv.D)* at the end of the file. Published files are found in the respective sub-folders of the *_published* folder.

[2i] Load Combinations

This is an inline table using the restructured text syntax.

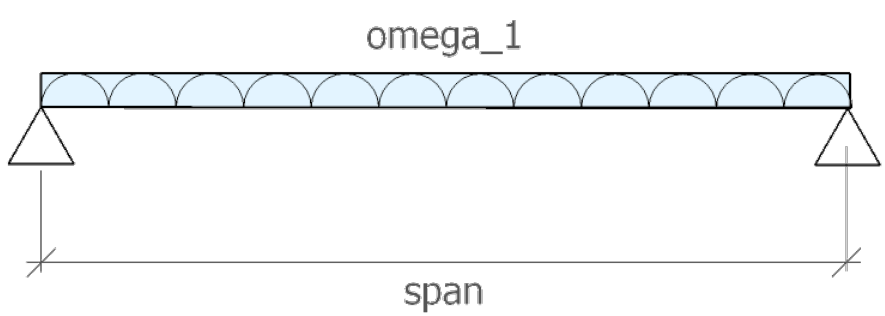
Table 1: ASCE 7-05 Load Effects

Equation No.	Load Combination
16-1	1.4(D+F)
16-2	1.2(D+F+T) + 1.6(L+H) + 0.5(Lr or S or R)
16-3	1.2(D+F+T) + 1.6(Lr or S or R) + (f1L or 0.8W)

When an inline table is in a TABLE block it produces the same output as above and also writes the table to a CSV file in the _stored folder.

Table 2: ASCE 7-05 Load Effects (saved as csv)

Equation No.	Load Combination
16-1	1.4(D+F)
16-2	1.2(D+F+T) + 1.6(L+H) + 0.5(Lr or S or R)
16-3	1.2(D+F+T) + 1.6(Lr or S or R) + (f1L or 0.8W)



Maximum Bending Stress

[Eq.1]

$$\sigma_1 = \frac{M_1}{S_1}$$

[3v] Loads and Geometry

Successive lines of value definiiions are formatted as a table. Variable values are defined with the define operator. The line tag [T] labels and numbers the table.

Table 3: Define Unit Loads

variable	value	[value]	description
D_1	3.80 psf	0.18 kPA	joists DL
D_2	2.10 psf	0.10 kPA	plywood DL
D_3	10.00 psf	0.48 kPA	partitions DL
D_4	3.00 klf	43.78 kN_m	fixed machinery DL
L_1	40.00 psf	1.92 kPA	ASCE7-O5 LL

The VALTABLE command reads variable values from the file in the _src folder. The text is used as the table title. The range specifies the starting and ending line to be read from the file (0:0 means all lines). The *num;non* parameter specifies whether theimported table is numbered.

Table 4: Beam Geometry [file: beam1.csv]

variable	value	[value]	description
spc_1	2.00 ft	0.61 m	beam spacing
spn_1	16.00 ft	4.88 m	beam span

Uniform Distributed Loads

[Eq.2]

```
dl_1 = 1.2*(D_4 + spc_1*(D_1 + D_2 + D_3))
```

spc_1	D_4	D_1	D_3	D_2
2.00 ft	3.00 klf	3.80 psf	10.00 psf	2.10 psf

dl_1	[dl_1]	reference
3.64 klf	53.09 kN_m	dead load: ASCE7-05 2.3.2

[Eq.3]

```
ll_1 = 1.6*L_1*spc_1
```

spc_1	L_1
2.00 ft	40.00 psf

ll_1	[ll_1]	reference
0.13 klf	1.87 kN_m	live load: ASCE7-05 2.3.2

[Eq.4]

```
omega_1 = dl_1 + ll_1
```

dl_1	ll_1
3.64 klf	128.00 ft·psf

omega_1	[omega_1]	reference
3.77 klf	54.96 kN_m	total load: ASCE7-05 2.3.2

[4v] Beam Response

Function Table 5: Beam functions **[file:** _src/sectprop.py **]**

Function	Docstring
rectsect(b, d)	section modulus of rectangle
rectinertia(b, d)	moment of inertia of rectangle
midspan_delta(ln, w, e, i)	mid-span deflection of simply supported beam with UDL

[Eq.5]

```
section_1 = rectsect(10*inch, 18*inch)
```

section_1	[section_1]
540.00 in3	8849.01 cm3

[Eq.6]

```
inertia_1 = rectinertia(10*inch, 18*inch)
```

inertia_1	[inertia_1]
4860.0 in4	202288.5 cm4

Bending Stress

[Eq.7]

```
                    2
      omega_1*spn_1
m_1 = --------------
            8
```

spn_1	omega_1
16.00 ft	3.77 klf

m_1	[m_1]	reference
120.52 ftkip	163.40 mkN	mid-span UDL moment

[Eq.8]

```
          m_1
fb_1 = ---------
       section_1
```

section_1	m_1
540.0 inch3	120.5 ft2·klf

fb_1	[fb_1]	reference
2678.2 lb_in2	18.5 MPA	bending stress

[Eq.9]

```
fb_1 < 20000*lb_in2
```

fb_1	20000*lb_in2	ratio	check	reference
2.68 ksi	20.00 ksi	0.133908	OK	stress ratio

Deflection

[Eq.10]

```
delta_1 = midspan_delta(spn_1, omega_1, 29000*ksi, inertia_1)
```

delta_1	[delta_1]
0.04 inch	1.00 mm